Dynamic Host configuration protocol (DHCP) :-

Dynamic assignment of IP addresses is desirable for several reasons :

1) IP addresses are assigned on demand.
2) Avoid manual IP configuration
3) support mobility of laptops.

→ DHCP is an extension of BOOTP.

→ DHCP supports temporary allocation ("leases") of IP addresses.

→ DHCP client can acquire all IP configuration parameters. (eg: subnet mask, gateway addres ...)

→ DHCP port number — 67 server, 68 client } DHCP uses UDP services

1) DHCP discover message → generated by client to find DHCP server.

2) DHCP offer message → The server response with IP address.

3) DHCP request message → in order to find out is there any host with same IP add in the n/w.

4) DHCP ACK message → The server make an entry with IP address of client bind IP to client

5) DHCP negative ACK →

6) DHCP decline → offered parameters are different or invalid client generates message to server

7) DHCP release - client send DHCP release packet to server for releasing space

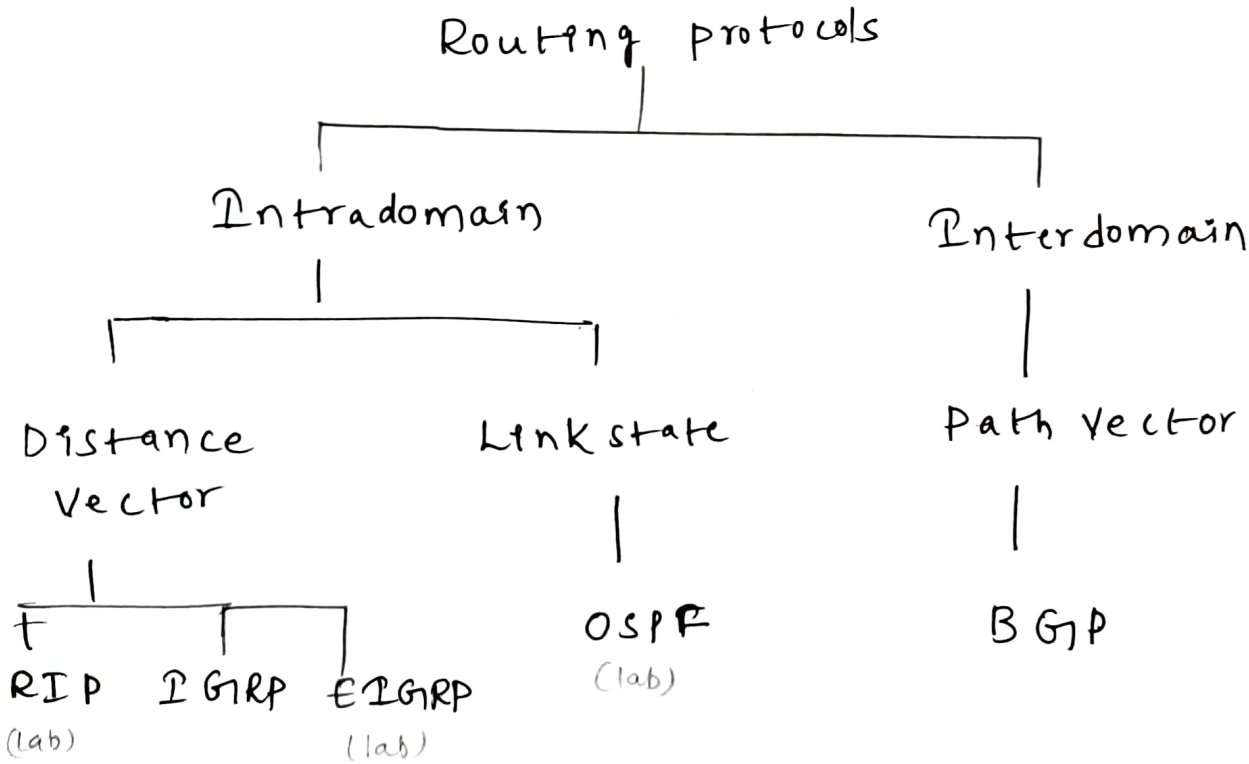
Routing is a process of selecting a path for traffic in a network or b/w across multiple networks.

→ An autonomous system (AS) is a group of networks and routers under the authority of a single administration.

→ Routing inside an autonomous system is referred to as Intradomain routing.

→ Routing between autonomous systems is referred to as interdomain routing.

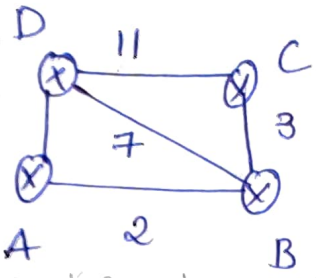
<u>popular routing protocols</u>



Distance Vector Routing

D Routing table

Destination	Distance	next hop
A	1	A
B	7	B
C	11	C
D	0	D



C Routing table

Destination	Distance	next hop
A	∞	—
B	3	B
C	0	C
D	11	D

A's Routing table

Destination	Distance	next hop
A	0	A
B	2	B
C	∞	—
D	1	D

B's routing table

Destination	Distance	next hop
A	2	A
B	0	B
C	3	C
D	7	D

→ Distance vector (array) is shared with neighbors.

→ Every node will get distance vectors from all its neighbors.

eg: A — {B, D}

B — {A, C, D}

C — {B, D}

D — {A, B, C}

At A

A receives distance vector from B and D.

→ A uses distance vectors from B and D to compute new routing table.

A previously	from B	from D
0	2	1
2	0	7
∞	3	11
1	7	0

New table @A

Dest	Dist	next hop
A	0	A
B	2	B
C	5	B
D	1	D

$A \to B = \min \begin{cases} \overset{2}{A \to B} \\ A \to D + D \to B \\ 1 + 7 = 8 \end{cases} \Rightarrow \min \begin{cases} 2 \\ 8 \end{cases} \Rightarrow 2$

$$A \to C = \min \begin{cases} \overset{1}{A \to D} + \overset{11}{D \rightsquigarrow C} \\ \underset{2}{A \to B} + \underset{3}{B \rightsquigarrow C} \end{cases}$$

$$= \min \begin{cases} 12 \\ 5 \end{cases} \Rightarrow 5 \quad (B)$$

$$A \to D = \min \begin{cases} \overset{1}{A \to D} + \overset{0}{D \rightsquigarrow D} \\ \underset{2}{A \to B} + \underset{7}{B \rightsquigarrow D} \end{cases}$$

$$\min \begin{cases} 1 \\ 9 \end{cases} \Rightarrow D$$

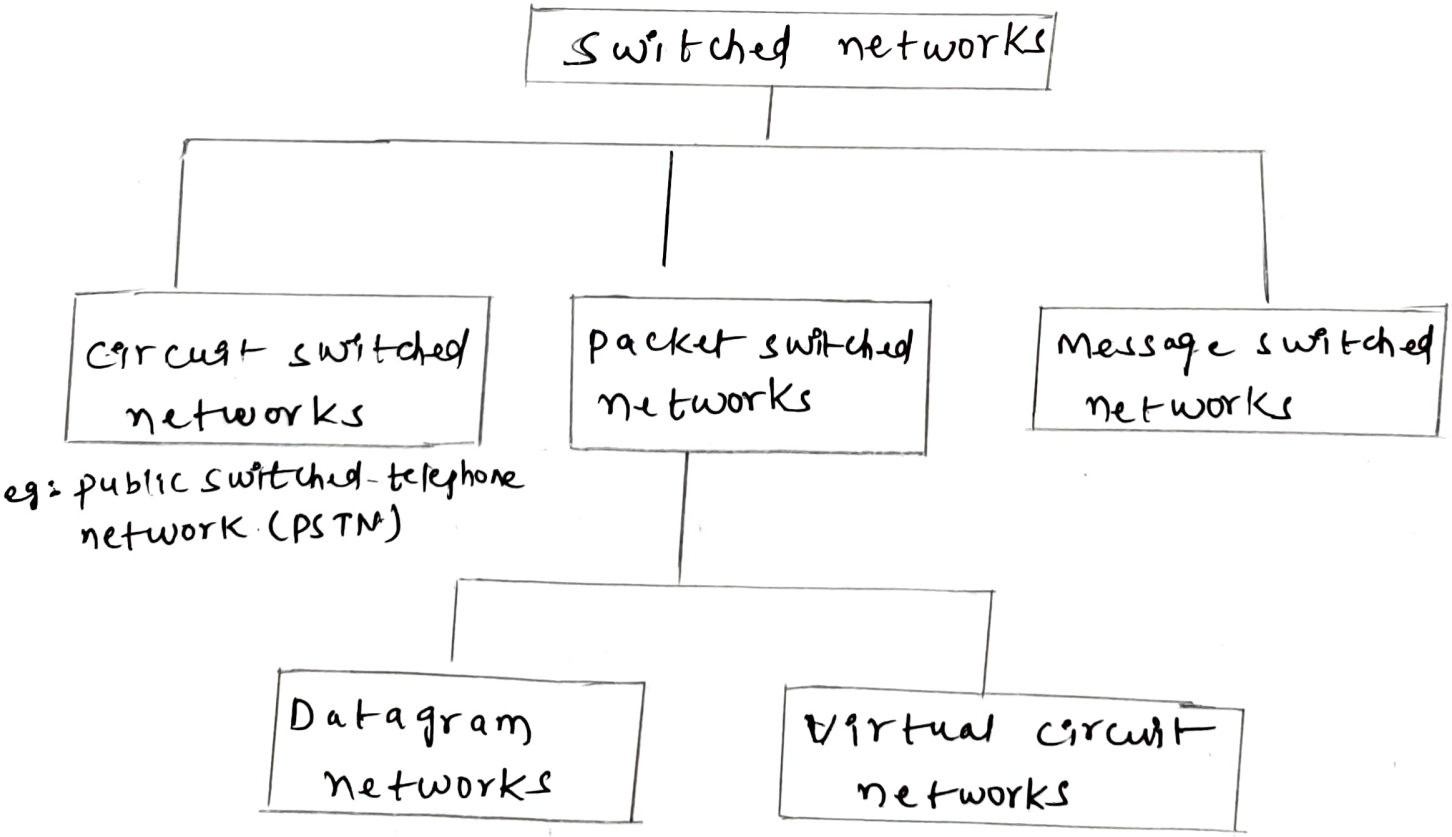
8m.

A switched network consists of a series of interlinked nodes, called switches. switches are devices capable of creating ~~creating~~ tempory connections b/w two or more devices linked to the switch.

→ Traditionally 3 methods of switching have been important. 1) circuit switching 2) packet switching 3) message switching.

→ we can divide today's n/w's into 3 broad categories.

Taxonomy of switched networks



Circuit switching : circuit switching takes place at physical layer.

communication via circuit switching implies that there is a dedicated communication path between two stations. circuit switching uses either of two technologies: the space division switch or time division switch.

→ The path is a connected sequence of links between network nodes.

→ on each physical link, a logical channel is dedicated to the connection. In circuit switching, the resources need to be reserved during the setup phase, the resources remain dedicated for entire duration of data transfer phase until teardown phase.

There are 3 important phases in circuit switching.

1) circuit establishment (or) setup phase
2) Data Transfer
3) circuit disconnect (or) teardown phase.

Circuit establishment

→ To establish an end-to-end connection before any transfer of data.

→ Some segments of the circuit may be a dedicated link, while some other segments may be shared.

Data Transfer

→ Transfer data is from source to destination

→ The data may be analog or digital, depending on the nature of the network.

→ The connection is generally full duplex.

Circuit disconnect :-

→ Terminate connection at the end of data transfer.

→ signals must be propagated to deallocate the dedicate the dedicated resources.

tual Circuit Networks

→ A preplanned route is established before any packets are sent.

→ call Request and call Accept packets are used to establish the connection.

→ Route is fixed for the duration of the logical connection (like circuit switching).

→ Each packet contains a virtual circuit identifier as well as data.

→ Each node on the route knows where to forward packets.

→ A clear Request packet issued by one of the two stations terminates the connection.

→ Route between stations is set up prior to data transfer.

→ A packet is buffered at each node, and queued for output over a line.

→ A data packet needs to ~~needs~~ carry only the virtual circuit identifier for effecting routing decisions.

→ Intermediate nodes take no routing decisions.

→ often provides sequencing and error control.

→ In virtual circuit Network three phases will be there. In setup phase the source and destination use their global addresses to help switches make table entries for the connection.

→ In teardown phase, the source and destination inform the switches to delete corresponding entry.

→ Data transfer occurs b/w these two phases.

Advantages:

1) Fixed bandwidth, guaranteed capacity (no conge

2) Low variance in end-to-end delay (constant delay).

Disadvantages:

1) circuit establishment and circuit disconnect introduces extra overhead and delay.

2) constant data rate from source to destination.

3) channel capacity is dedicated for the duration of the connection, even if no data is being transffered.

4) other users cannot use it even if free of traffic.

5) Not suitable for bursty data.

Packet switching:

→ In packet switching, there is no resource allocation for a packet. This means that there is no reserved bandwidth on the links, and there is no scheduled processing time for each packet. Resources are allocated on demand.

→ Data are transmitted in short packets (few kbytes).

→ A longer message is broken up into a series of packets.

→ Every packet contains some control information in its header (required for routing and other purposes).

→ Two approaches are commonly used for handling these packets: 1) Virtual circuit 2) Datagram.

→ In packet switched network, there is no resource reservation; resources are allocated on demand.

Datagram approach

→ In this approach ther is no setup or tear down phases.

→ Each packet is treated independently, with no reference to packets that have gone before.

→ Every intermediate node has to take routing decisions.

→ Every packet contains source and destination addresses.

→ Intermediate nodes maintains routing tables.

Advantages :

→ call setup phase is avoided. (for transmission of a few packets, datagram will be ~~transf~~ faster).

→ Because it is more primitive, it is more flexible.

→ congestion / failed link can be avoided (more reliable).

Problems :

→ packets may be delivered out of order.

→ If a node crashes momentarily, all of its queued packets are lost.